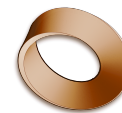


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Date: _____

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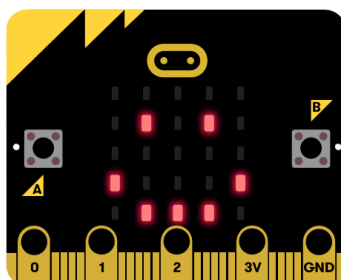


1.2 – Algorithms 1 — making the micro:bit decide

Unit 1: Robotics & Control

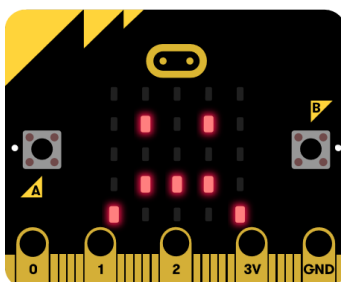
Last lesson your micro:bit did the same thing every time.
Today it starts **deciding**.

1 Show a happy face when you shake it.



on shake + show icon

2 Show a sad face when you press A.



on button A pressed + show icon

3 Show a heart when you press B *four times*.

No picture — you have to make this one appear.

You cannot do it yet. Try anyway, then work out *why* it will not go.

micro:bit images © BBC micro:bit Educational Foundation, from the *Introduction to Computer Science* course. Used for classroom teaching.

Why number 3 is different

Challenges 1 and 2 are **events**. Something happens, the micro:bit answers.

Number 3 is not. The micro:bit has to **remember** how many times you pressed B, then **check** that number.

Remembering needs a **variable**. Checking needs a **condition**.

A variable is a box with a name on it

You put a number in the box. You can change it whenever you like.

B Clicks

0

so far, B has been pressed **0** times

Three things you do with a box:

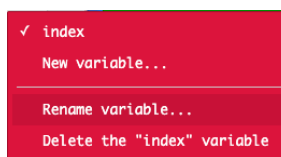
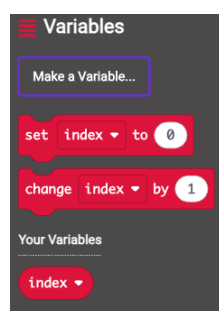
Variables → **Make a Variable** build the box, call it B Clicks

set B Clicks to 0

put 0 in the box

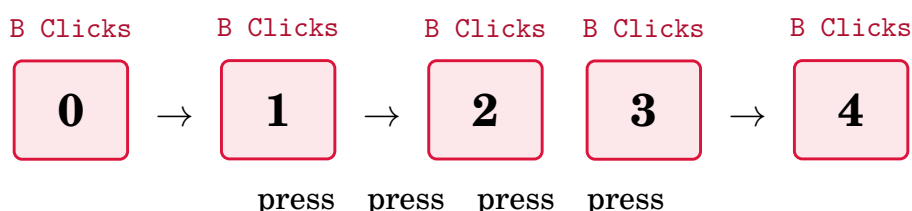
change B Clicks by 1

add one to what is in the box



The **Variables** drawer, and the menu you get after **Make a Variable**.
index is only the name MakeCode starts with — yours will say B Clicks.

So every press of B moves the number along:



A condition is a question with only two answers

Yes, or no. Nothing else. This is the question we want to ask:

B Clicks = 4

“Is what is in the box equal to 4?”



From the **Logic** drawer — the if block, and the = block that drops into its gap.

Ask it after every press, and the answer changes:

press	B Clicks = 4?	
1st	no	keep going
2nd	no	keep going
3rd	no	keep going
4th	YES	show the heart

The block that asks the question is `if ...then`.

Whatever you put *inside* it only happens when the answer is **yes**.

Watch out

`set B Clicks to 0` **puts** a number in the box.

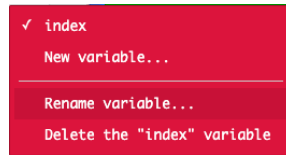
`B Clicks = 4` **asks** about the number.

They both use =, and they do completely different jobs.

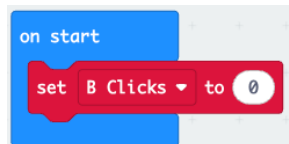
Now build challenge 3

Follow these in order. **Step 6 is yours to work out.**

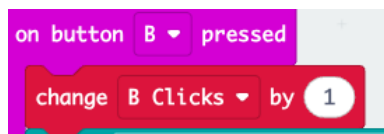
1 Make a variable. Call it B Clicks.



2 In `on start`, put `set B Clicks to 0`.



3 Start a *new* stack: `on button B pressed`, with `change B Clicks by 1` inside.

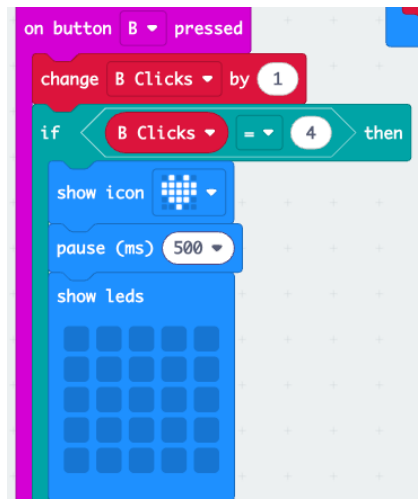


4 Under that, add `if ...then`. Drag `B Clicks = 4` into the gap.



5 *Inside* the if: show the heart, pause, then clear the screen.

Everything you add must sit **inside** the teal jaws, lined up under `show icon` — not underneath the if block.



6 Press B six times. What goes wrong? Fix it.

The heart appears once, and then never again. Work out why, and add **one more block** to fix it. It goes inside the if.

Word bank

input	information going <i>in</i> — a button, a shake
output	what comes <i>out</i> — the LEDs, a sound
event	something that happens, and starts code running
variable	a named box holding a value you can change
condition	a question with a yes or no answer
algorithm	the steps, in order, that solve the problem

HOMEWORK — The Shake Counter

Put this sheet away first, and try it from memory. Page 4 will hand you the answer, which is not the point. Look back only if you get properly stuck.

Same idea as challenge 3. This time you build it on your own.

1. Make a variable called Shakes.
2. Every **shake** adds 1 to it.
3. On the **3rd** shake, show an icon — you pick which one.
4. Then it starts again from zero, ready for the next three.

Share → Publish project → submit the URL to **Google Classroom**.

Learn the six words — you will be asked next lesson.